

URBADAPT-HEAT — Urban Heat Adaptation Framework

Probabilistic, city-level cost–benefit analysis of heat-mortality adaptation



`configs/{city}.yaml` · city-agnostic `template/`

Per-city parameters & policy — hazard track, income source, EWS mode, costs. One template driven by the CITY variable → drives all 10 notebooks identically



`cityheat/` `engine` · `scripts/`

Shared scientific code + reproducible generator & Drive data manifests

Data retrieval and processing modules

NB1 Setup & Loading

City configuration (YAML)
LCZ, GVI, UrbClim T2M (daily mean + max)
Population & demographics · Drive manifests

NB2 Harmonised Grids

Harmonise to 100 m grid (EPSG:3035)
FUA & administrative masks
Dynamic vulnerability layers

Climate Risk Model

NB3 Hazard & Exposure

CLIMADA hazard & exposure objects
Daily T2M (2020–2050)
Dynamic, time-varying exposure (age-stratified population, by scenario & year)

→ Temperature hazard

NB4 Impact Functions

Masselot age-specific mortality ERFs (log-linear tail extrapolation)
Burke sensitivity + cold-city Track-A/B

→ Impact functions

↓
Age-resolved mortality risk (pre-adaptation)

Adaptation Strategies — Each computes its own costs & benefits

NB5 AC Policy

PRIVATE

AC coverage scenarios (SSP2)
Within-city 🏠 **income emulator**
Efficacy-adjusted mortality reduction
CAPEX + O&M + electricity

→ Vulnerability ↓ reduces mortality given T

⚡ **Side effect:** Waste heat → Hazard ↑ (+ΔT)
Exports *marginal deaths per °C* for penalty calc

→ GROSS benefits + Costs + Marginal

NB6 Early Warning

PUBLIC

Heat–health warning system
Three efficacy cases:
marginal / counterfactual / intermediate
Infrastructure + operational costs
25-year benefit interpolation

→ Vulnerability ↓ behavioural response

→ Benefits + Costs (25y PV)

NB7 Urban Trees

PUBLIC

GVI cooling coefficients (ΔGVI→ΔLST, by LCZ)
🏠 **LST→T2M emulator** bridge (142 cities)
Tree-maturity ramp · regional / equity allocation
CAPEX + O&M costs

→ Hazard ↓ ΔGVI→ΔLST→emulator→ΔT2M

🌱 **Co-benefit:** Greening → AC electricity ↓
Runtime-calibrated from JJA daily-max T2M

→ Benefits + Costs + Equity metrics

↓
Policy-specific costs, benefits & feedback parameters

Economics, Uncertainty & Synthesis — Integration & comparison

NB8 CBA & Optimisation

Integrate NB5 + NB6 + NB7
25-year NPV (3% central discount)
Budget-constrained optimisation
Public vs private cost split

Feedback adjustments:

Waste-heat penalty (marginal deaths/°C):
NET AC = GROSS – penalty
Vegetation–AC electricity co-benefit

→ Net benefits, NPV & cost-effectiveness

NB9 Uncertainty

Monte-Carlo sampling over:
climate scenario & GCM
impact-function family & T-ref
adaptation efficacy
costs & discount rate
Propagated through the full pipeline

→ Probabilistic deaths avoided & CE

NB10 Framework Summary

Cross-pathway & cross-city synthesis
Comparative figures & tables
Ranking of adaptation options
by cost-effectiveness & equity
Reproducible per-city report

→ Decision-ready comparison